PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	**IMRAN ALI HAIDER**
Program/Batch	B084-Networkwalks
Date	12 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (networkwalks.com) with written permission (Letter of Authorization) 2. My own Kali Linux lab machine (localhost, 127.0.0.1)
Permission secured from client?	Yes: written Letter of Authorization from Networkwalks
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** Not performed in this report

1. Liability Disclaimer

I have performed these activities only on the systems and devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1) and scanning my own lab machine with Nmap and Zenmap (W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from gathering public information to discovering hosts and open services. It is the Week 2 part of my ongoing internship program at Networkwalks.

All commands were run in a Kali Linux virtual machine (VirtualBox). The testing stayed within the scope of the Letter of Authorization: the public website networkwalks.com and its DNS records, and my own machine. Only passive and light information-gathering tools were used. No exploitation, brute-force or denial-of-service activity was performed. Every step below includes the exact command used, the result I observed, a reference to the screenshot evidence in Section 8 and a short note on why the finding matters.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux (VirtualBox VM)	Operating system used for all reconnaissance and scanning activities
whois	Find domain registration details (registrar, dates, name servers)
nslookup	Query DNS for the IP address and the A, NS, MX and TXT records

Tool	Purpose
whatweb	Fingerprint web technologies (server, CMS, plugins, IP)
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, TXT, SRV)
Nmap and Zenmap (Nmap GUI)	Host discovery, port and service detection, OS detection and topology view on my own machine

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the `networkwalks.com` domain using six Kali Linux tools: **whois, nslookup, whatweb, curl, wafw00f and dnsrecon**. Each tool collected a different type of information about the target.

whois (`whois networkwalks.com`). The domain is registered through GoDaddy.com, LLC. It was created on 6 November 2019, was last updated on 12 November 2025 and expires on 6 November 2027. The name servers are NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM, so DNS is hosted by HostGator. All four client lock statuses (delete, renew, transfer and update prohibited) are set, and DNSSEC is shown as unsigned. The output ended with a notice that the registry whois service is being retired in favour of RDAP, and a rate-limit message after repeated queries, so I did not query it again.

nslookup (`nslookup networkwalks.com`, then `-type=NS`, `-type=MX` and `-type=TXT`). The domain resolves to **192.232.216.135**. The name servers are ns6135.hostgator.com and ns6136.hostgator.com. Mail is handled by mail.networkwalks.com (priority 0). The TXT records contain a Google site verification token and the SPF policy `v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`. The answers were non-authoritative because they came from the Google resolver 8.8.8.8.

whatweb (`whatweb networkwalks.com`). HTTP requests are redirected (301) to HTTPS. The site runs on Apache and WordPress 7.1.4 with the WordPress Download Manager plugin 3.3.76. It also uses jQuery 3.7.1 and Google Tag Manager, exposes the public email address info@networkwalks.com, and sets an HttpOnly cookie. Headers such as permissions-policy and referrer-policy are present. WhatWeb also reported a Bootstrap version equal to the WordPress version, which is probably a misdetection, so I do not treat it as a finding.

curl (`curl -I https://networkwalks.com`). The server answered HTTP/2 200 and identifies itself only as Apache (no version number). The response included Permissions-Policy and Referrer-Policy headers, a cookie with the Secure and HttpOnly flags, caching headers (x-nginx-cache, x-endurance-cache-level) and a Link header exposing the WordPress REST API endpoint `/wp-json/`. I did not see Strict-Transport-Security, Content-Security-Policy, X-Frame-Options or X-Content-Type-Options in this response.

wafw00f (`wafw00f https://networkwalks.com`). The tool detected that the site is behind a **ModSecurity (SpiderLabs)** Web Application Firewall, using 2 requests.

dnsrecon (`dnsrecon -d networkwalks.com -t std`). Standard enumeration found 8 records. It listed the SOA and NS records (ns6135.hostgator.com at 50.87.144.87 and ns6136.hostgator.com at 192.232.216.131), the MX record mail.networkwalks.com and the A record 192.232.216.135 (so web and mail share one IP), the two TXT records, and eight _autodiscover._tcp SRV records pointing to cpanelemaildiscovery.cpanel.net. Both name servers disclosed the DNS software version "9.16.23-RH". The tool reported no answer to the DNSSEC query. My first attempt used a full URL and failed with a resolve error; I re-ran it with the bare domain name. I did not use any brute-force option.

4.2 Network Scanning with Nmap and Zenmap

For the second activity I scanned my own Kali Linux lab machine (`127.0.0.1`) with Nmap in the terminal and with **Zenmap**. This is covered by the "tester's own local area network" item in the Letter

of Authorization. I started the Apache web server on the machine first so that the scan had a service to find, and stopped it again afterwards. The scan covered a single host, so there are no other hosts or MAC addresses to report (a loopback address has no MAC).

Host discovery. A ping scan (`nmap -sn 127.0.0.1`) confirmed that the host is up (1 IP address scanned, 1 host up). I repeated it in Zenmap with the Ping scan profile.

Port and service detection. `sudo nmap -sV 127.0.0.1` scanned 1000 ports and found 999 closed and one open: **80/tcp open http, Apache httpd 2.4.69 (Debian)**.

Aggressive scan and OS detection. `sudo nmap -A 127.0.0.1` returned the default Apache page title ("Apache2 Debian Default Page: It works") and the server header Apache/2.4.69 (Debian). The operating system guess was Linux 5.0 to 6.2 (97%), with no exact match, and the network distance was 0 hops.

Zenmap Intense scan. In Zenmap I set the target to 127.0.0.1 and chose the Intense scan profile (`nmap -T4 -A -v 127.0.0.1`). It completed in 18.75 seconds. The Ports/Hosts tab lists port 80, tcp, open, http, Apache httpd 2.4.69. Host Details shows 1 open port, 0 filtered, 999 closed and 1000 scanned, IPv4 127.0.0.1, hostname localhost, and the OS guess Linux 5.0 - 6.2. The Topology tab shows a single node, localhost. The output also noted TCP sequence prediction difficulty 256 and an IP ID sequence of all zeros.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	CMS and plugin versions exposed	WhatWeb identified WordPress 7.1.4 and WordPress Download Manager 3.3.76 (meta generator tag)	Attackers can look up known issues for the exact versions in use	• **Medium**
2	Missing browser security headers	curl -I response showed Permissions-Policy and Referrer-Policy, but no HSTS, CSP, X-Frame-Options or X-Content-Type-Options header; /wp-json/ endpoint advertised in the Link header	Weaker browser-side protection against clickjacking, content sniffing and downgrade attacks	• **Medium**
3	DNS and mail configuration details exposed	dnsrecon found BIND version 9.16.23-RH on both name servers, cPanel SRV records, and an SPF record ending in ~all (soft fail)	Helps profile the infrastructure; soft-fail SPF makes email spoofing easier	• **Medium**
4	Server IP and hosting identifiable	nslookup resolved the site to 192.232.216.135; web, mail and name servers sit on HostGator shared hosting	Shows where the service is hosted; shared hosting means shared exposure	• **Low**
5	WAF technology identifiable	wafw00f identified ModSecurity (SpiderLabs)	Reveals part of the security architecture (the WAF itself is a positive control)	• **Low**
6	DNSSEC not enabled	whois showed DNSSEC: unsigned; dnsrecon got no answer for the DNSSEC query	DNS answers are not cryptographically validated	• **Low**
7	Local web service with version banner	Nmap found 80/tcp open on 127.0.0.1 running Apache httpd 2.4.69 (Debian) with the default "It works" page	Unneeded service and version banner increase attack surface on a host	• **Low**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. These two modules involved information gathering and host discovery only. No exploitation or vulnerability validation was performed. A software version, IP address or DNS record does not by itself mean that a system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability. No risk is rated Critical because nothing in this report shows direct exploitability.

6. Recommendations

1. **Reduce technology disclosure.** Remove the WordPress generator meta tag and review what CMS, plugin and version information is publicly visible.
2. **Keep software updated.** Keep WordPress core, plugins (including WordPress Download Manager) and libraries up to date and review them against current security advisories.
3. **Add missing security headers.** Add Strict-Transport-Security, Content-Security-Policy, X-Frame-Options and X-Content-Type-Options, and add the SameSite attribute to cookies.
4. **Harden email authentication.** Move the SPF policy from ~all to -all once all legitimate senders are confirmed, and publish DKIM and DMARC records.
5. **Harden DNS.** Hide the BIND version string on the name servers, enable DNSSEC if the DNS host supports it, and review DNS records regularly so that only required information is exposed.
6. **Properly configure and monitor the WAF.** Keep ModSecurity enabled, its rules updated, and its logs monitored.
7. **Stop unneeded services and hide banners.** Stop services that are not needed, set ServerTokens Prod and ServerSignature Off on Apache, remove the default page, and use a firewall to allow only required ports.
8. **Perform regular internal network discovery.** Scan your own networks periodically to identify active devices, investigate unknown devices, and keep network documentation up to date.
9. **Perform security testing only with authorization.** Reconnaissance and scanning should only be performed against systems and networks where written authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how whois provides registration details, nslookup resolves DNS records, whatweb identifies web technologies, curl shows HTTP headers, wafw00f identifies a WAF, and dnsrecon enumerates DNS records. Together they showed that WordPress and plugin versions, DNS software details and missing security headers are visible to anyone, while controls such as HTTPS redirection, a WAF, HttpOnly cookies and domain locks are in place.

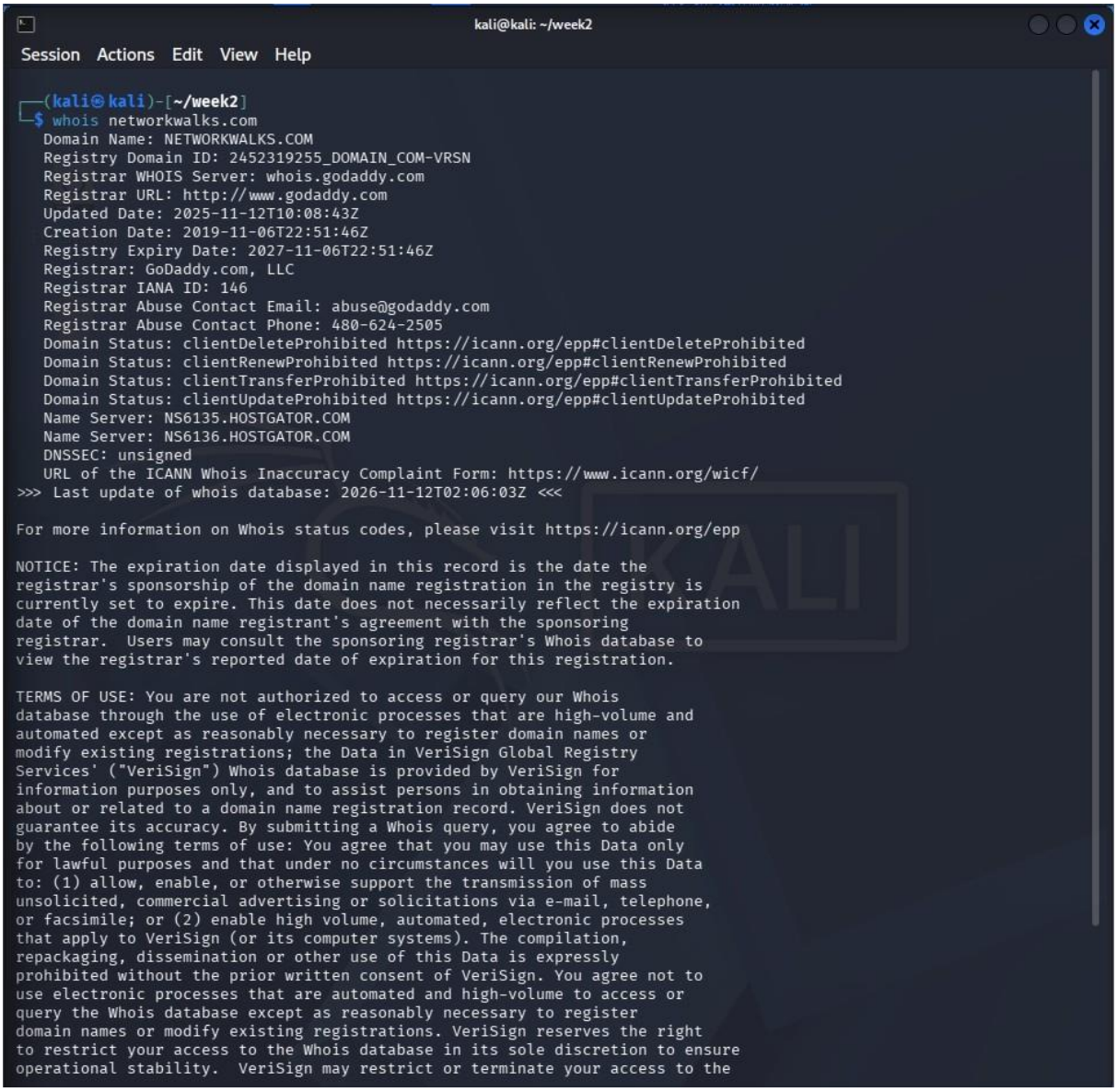
In the network scanning activity, I used Nmap and Zenmap to discover a host, detect its open port and service version, guess its operating system and view its topology. This scan was limited to a single machine (my own Kali VM at 127.0.0.1), so it did not show multiple hosts or MAC addresses. A future scan of a full subnet with several devices would give a more complete picture of a network.

The exercises showed me that information gathering is an important part of cybersecurity, that findings should be documented clearly with evidence, and that reconnaissance and scanning must always stay within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

The screenshots below are the evidence for the activities described in Section 4. Each one shows the command that was run and its output.

Screenshot 1: whois networkwalks.com



Screenshot 2: nslookup networkwalks.com (A record)



Screenshot 3: nslookup -type=NS networkwalks.com



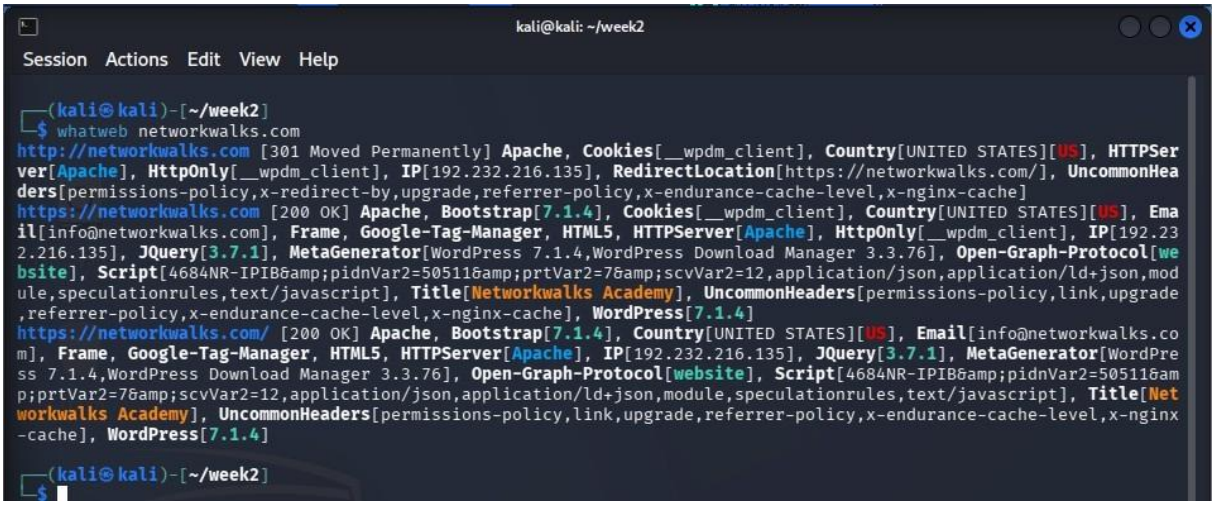
Screenshot 4: nslookup -type=MX networkwalks.com



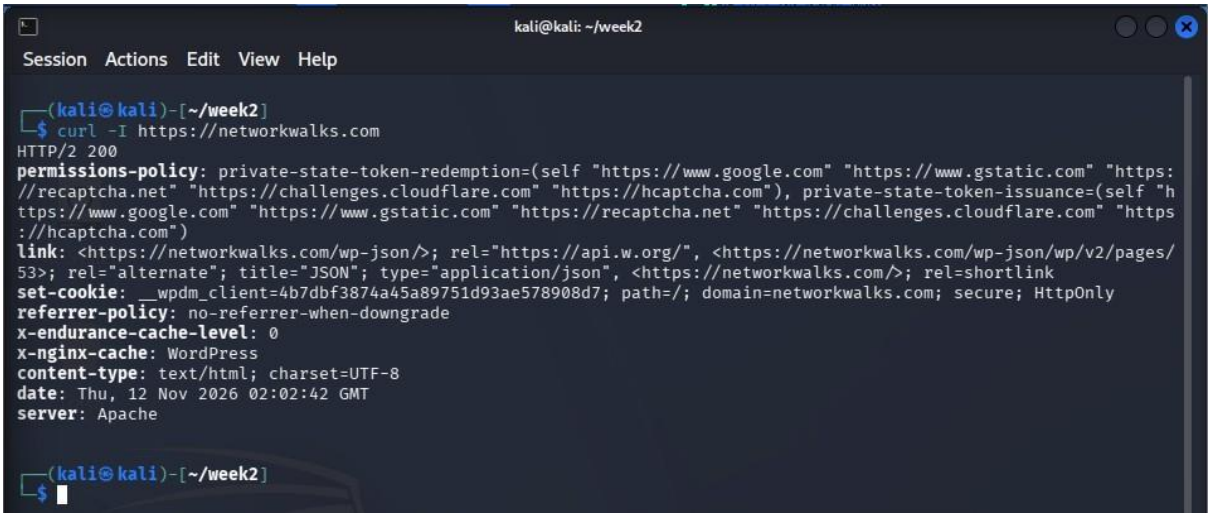
Screenshot 5: nslookup -type=TXT networkwalks.com



Screenshot 6: whatweb networkwalks.com



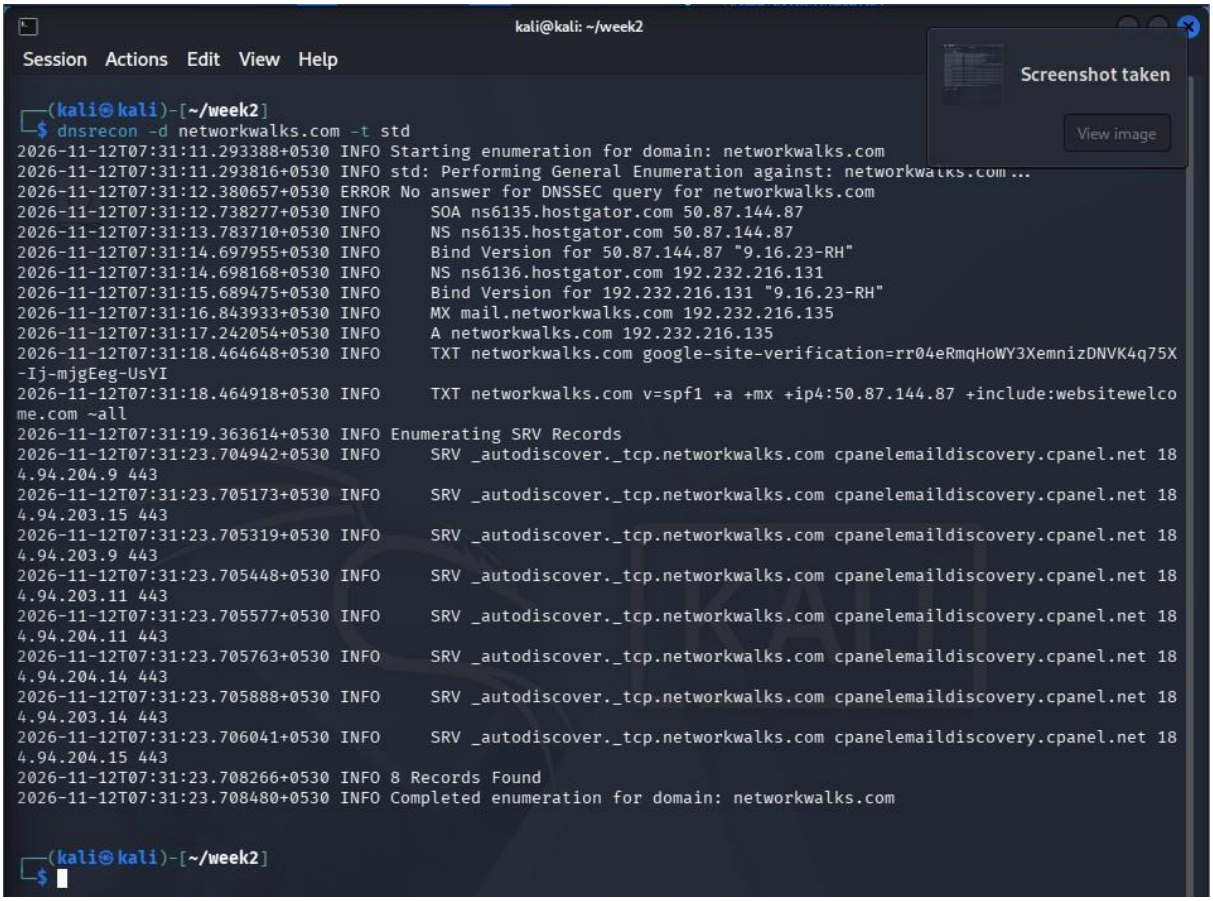
Screenshot 7: curl -I https://networkwalks.com



Screenshot 8: wafw00f https://networkwalks.com



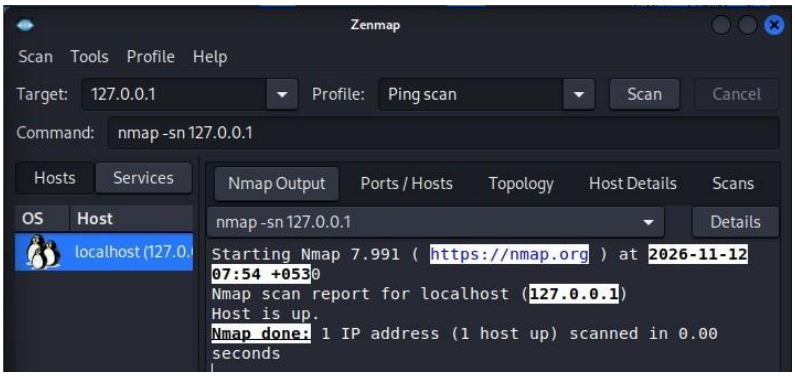
Screenshot 9: dnsrecon -d networkwalks.com -t std



Screenshot 10: nmap -sn 127.0.0.1



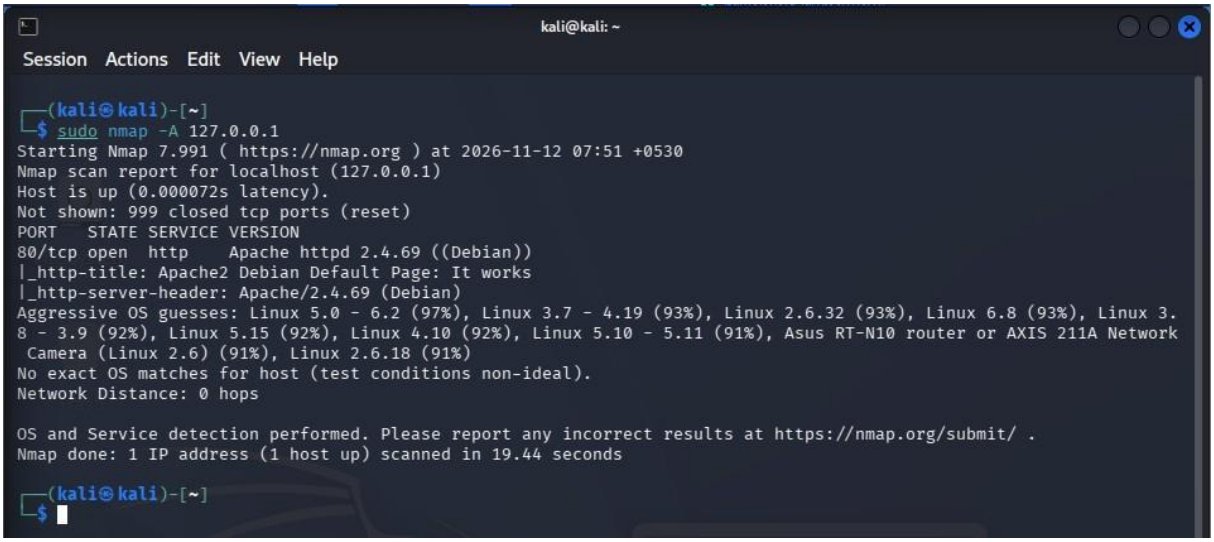
Screenshot 11: Zenmap Ping scan profile (nmap -sn 127.0.0.1)



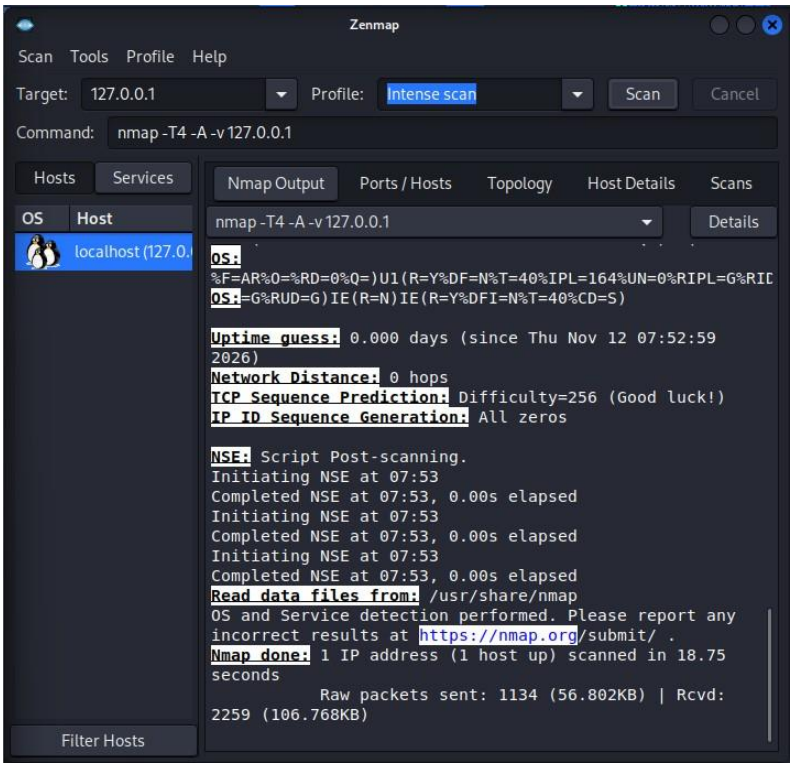
Screenshot 12: sudo nmap -sV 127.0.0.1



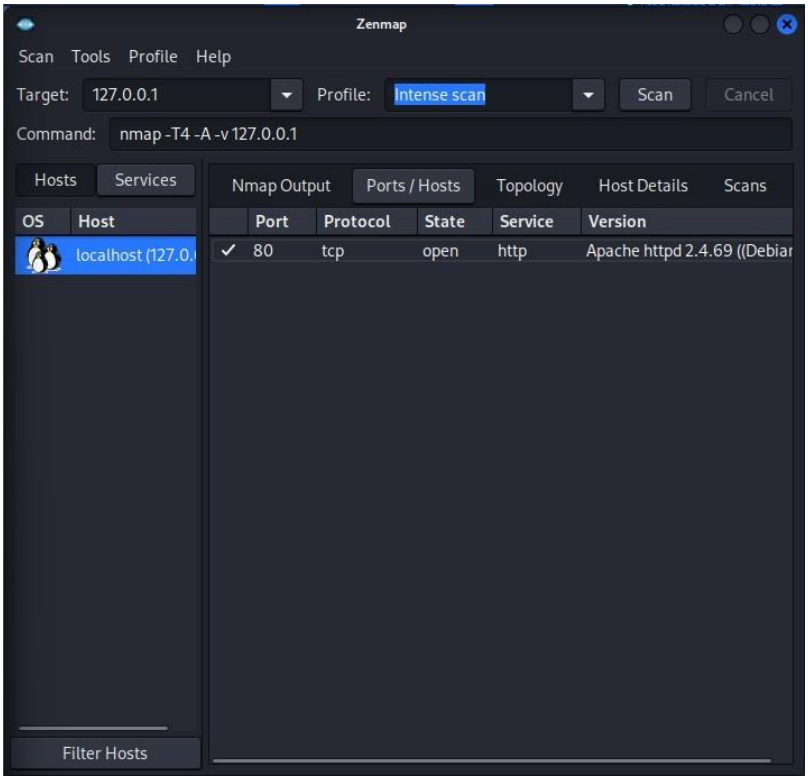
Screenshot 13: sudo nmap -A 127.0.0.1



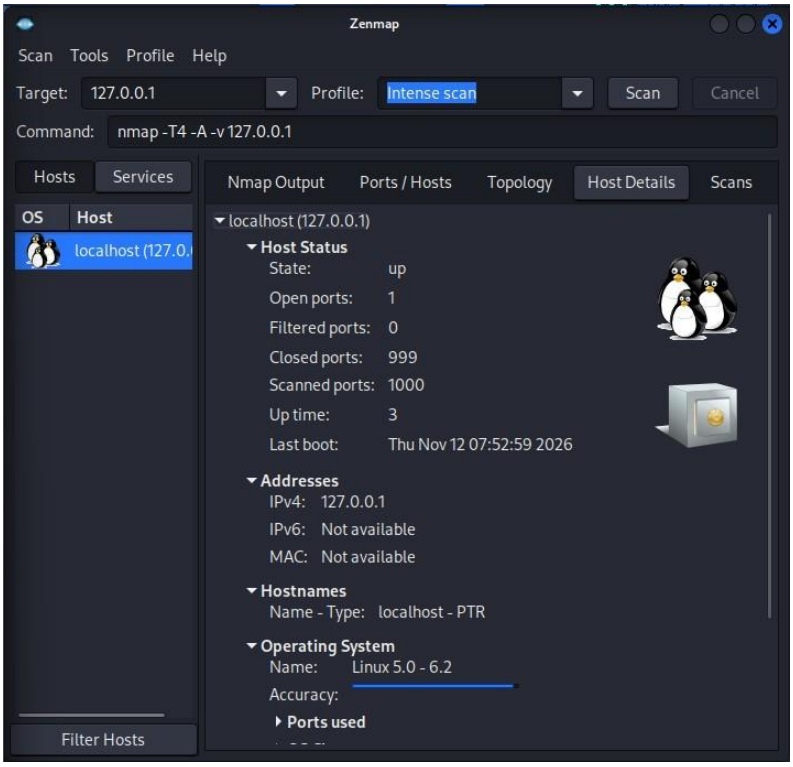
Screenshot 14: Zenmap Intense scan, Nmap Output tab



Screenshot 15: Zenmap Ports / Hosts tab



Screenshot 16: Zenmap Host Details tab



Screenshot 17: Zenmap Topology tab

